



Modelling Financial Markets

Chapter 20 (special chapter): Path Signatures and Realised Volatility — a Research Case Study



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Acronyms used in this material (1/2)

- ▣ AAPL: Apple Inc. (Nasdaq ticker)
- ▣ ACM: Association for Computing Machinery
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ AUD: Australian Dollar (ISO 4217 code)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BIC: Bayesian Information Criterion
- ▣ BPQ: Bollerslev–Patton–Quaedvlieg (insanity filter of the HARQ paper)
- ▣ CAD: Canadian Dollar (ISO 4217 code)
- ▣ CL: Crude oil (WTI light sweet) futures, NYMEX ticker
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DM: Diebold–Mariano (test)
- ▣ ES: E-mini S&P 500 futures (CME ticker)
- ▣ EURUSD: Euro / US dollar exchange rate (FX pair code)
- ▣ FDR: False Discovery Rate
- ▣ FWER: Family-Wise Error Rate
- ▣ FX: Foreign eXchange
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GC: Gold futures, COMEX ticker
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HARQ: HAR model with Realised Quarticity
- ▣ JPM: JPMorgan Chase & Co. (NYSE ticker)
- ▣ KDD: Knowledge Discovery and Data Mining (ACM SIGKDD conference)

Acronyms used in this material (2/2)

- LARS: Least-Angle RegreSsion (algorithm)
- LASSO: Least Absolute Shrinkage and Selection Operator
- MCS: Model Confidence Set
- MSE: Mean Squared Error
- NG: Natural Gas futures, NYMEX ticker
- NVDA: NVIDIA Corporation (Nasdaq ticker)
- OLS: Ordinary Least Squares
- PDE: Partial Differential Equation
- PRDS: Positive Regression Dependence on a Subset (of p-values)
- QLIKE: Quasi-LIKelihood (loss function)
- RQ: Realised Quarticity
- RV: Realised Variance
- SHAR: Semivariance HAR model
- TSLA: Tesla, Inc. (Nasdaq ticker)
- US: United States
- USDJPY: US dollar / Japanese yen exchange rate (FX pair code)
- VOLARE: VOLatility Archive for Realized Estimates (database)
- XOM: Exxon Mobil Corporation (NYSE ticker)

Chapter Route and Outcomes

Route of this chapter

- A research case study: the method of [Gu et al. \(2024\)](#), published for freight rates at Amazon, transferred to daily realised variance of 50 VOLARE assets [Cipollini et al. \(2026\)](#)
- Part I: realised measures, the data, the HAR benchmarks, path signatures
- Part II: the paper of Gu et al., a design fixed before the results, results, robustness, one pass of the AI-assisted discovery loop
- Implementation detail sits in the [Appendix](#) at the end

After this chapter you can

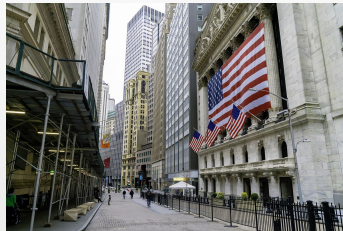
- compute the signature of a piecewise-linear path and read its Lévy area
- build signature features and kernel weights for a volatility forecast
- compare forecasts on many assets with DM tests, model confidence sets and Holm/BH corrections
- pre-register a method transfer and report a null result with its mechanism

Why Forecast Realised Variance?

- Volatility is an input to almost every risk measure in this course
 - ▶ Tail-risk measures ([Chapter 7](#)), option prices ([Chapter 12](#)), portfolio weights ([Chapter 4](#))
- Intraday data make daily variance almost observable: realised variance (RV, [Chapter 9](#))
 - ▶ So volatility forecasting becomes a regression problem with a noisy but model-free target
- The benchmark is hard to beat: the HAR model [Corsi \(2009\)](#) ([Chapter 9](#))
 - ▶ Three averages of past RV; linear; estimated by OLS in a second
- Any new method must beat HAR out of sample, on many assets, with a proper test

The Research Question

- Gu et al. describe a recent window of a series as a **path** and summarise it by its **signature**
 - ▶ Signature features enter a sparse linear regression (two-step LASSO)
 - ▶ Past periods whose path looks like today's get more weight (a signature kernel)
- They report a more than fivefold accuracy gain over an industry forecast of freight rates
 - ▶ Question: does the same recipe improve on HAR-type models for realised variance?
- Two separate ingredients to test: the signature features and the kernel weights



Wall Street and the New York Stock Exchange, 25
March 2020

Photo: [Billie Grace Ward \(2020\)](#); CC BY 2.0; [Wikimedia Commons](#)

Realised Variance and Quadratic Variation (1/2)

- The price model: the log price moves by a drift, a diffusion term and jumps

$$dp_t = \mu_t dt + \sigma_t dW_t + dJ_t$$

- ▶ p_t : log price at time t (time in days); μ_t : drift; σ_t : spot volatility; W_t : Brownian motion; J_t : pure-jump process, $\Delta J_s =$ jump size at time s

- Realised variance: the sum of squared intraday returns of day t [Barndorff-Nielsen & Shephard \(2002\)](#)

$$RV_t = \sum_{i=1}^M r_{t,i}^2 \xrightarrow{M \rightarrow \infty} \int_{t-1}^t \sigma_s^2 ds + \sum_{t-1 < s \leq t} (\Delta J_s)^2$$

- ▶ M : number of intraday returns in day t (for a US stock, 78 five-minute returns in the 6.5-hour session); $r_{t,i}$: the i -th intraday log return of day t
- ▶ The limit is the quadratic variation of day t : the integrated variance plus the sum of squared jumps

Realised Variance and Quadratic Variation (2/2)

- Without jumps and noise, on an equally spaced grid, the limit is the integrated variance; the error is mixed Normal

$$\sqrt{M}(\text{RV}_t - \text{IV}_t) \Rightarrow \text{MN}(0, 2\text{IQ}_t), \quad \text{IV}_t = \int_{t-1}^t \sigma_s^2 ds, \quad \text{IQ}_t = \int_{t-1}^t \sigma_s^4 ds$$

- ▶ IV_t : integrated variance, the true variance of day t ; IQ_t : integrated quarticity; $\Rightarrow$: convergence in distribution
- ▶ $\text{MN}(0, V)$: mixed Normal, a Normal with a random variance V ; given the volatility path, the error is Normal with variance $2\text{IQ}_t/M$
- Without jumps and noise, on an equally spaced grid, the limit is the integrated variance; the error is mixed Normal
- ▶ This is the measurement error that HARQ corrects for (next section)
- Microstructure noise makes very fine sampling biased; five-minute returns are the standard compromise ([Chapter 9](#))
- [Chapter 9](#) derives these results; they are used here as given

The Realised Measures in VOLARE (1/2)

- ▣ Semivariances Barndorff-Nielsen, Kinnebrock & Shephard (2010): RV split by the sign of the intraday returns

$$RV_t^+ = \sum_{i=1}^M r_{t,i}^2 1\{r_{t,i} > 0\}, \quad RV_t^- = \sum_{i=1}^M r_{t,i}^2 1\{r_{t,i} < 0\}$$

- ▶ $1\{\cdot\}$: indicator, 1 if the condition holds and 0 otherwise; RV_t^+ : variance from up-moves, RV_t^- : variance from down-moves
- ▶ They add up: $RV_t = RV_t^+ + RV_t^-$ (holds in VOLARE to machine precision)
- ▣ Realised quarticity: an estimate of IQ_t , the scale of the measurement error of RV

$$RQ_t = \frac{M}{3} \sum_{i=1}^M r_{t,i}^4 \rightarrow IQ_t$$

- ▶ The factor $M/3$: a Normal return with variance s^2 has fourth moment $3s^4$, and each five-minute variance is about IV_t/M
- ▶ The limit needs a continuous, noise-free price and an equally spaced grid; a jump ΔJ adds about $M(\Delta J)^4/3$, so raw RQ is jump-sensitive

The Realised Measures in VOLARE (2/2)

- Bipower variation (robust to jumps), realised kernel (robust to noise) [Barndorff-Nielsen et al. \(2008\)](#), range-based estimators
- Target of this chapter: five-minute RV
 - ▶ Across 400 assets, little beats five-minute RV as a proxy of daily variance [Liu, Patton & Sheppard \(2015\)](#)
 - ▶ Its semivariances and quarticity are on the same grid, so SHAR and HARQ use consistent inputs

Targets and Losses (1/2): the Target

- The target at horizon h is the average RV over the next h days

$$RV_{t+1:t+h} = \frac{1}{h} \sum_{j=1}^h RV_{t+j}, \quad h \in \{1, 5, 22\}$$

- ▶ t : the forecast origin (the last day observed); $h = 1, 5, 22$: one day, one week, one month of trading days

- Log models forecast $y = \ln RV_{t+1:t+h}$ and return to levels with a variance correction

$$F_t = \exp(\hat{y}_t + \hat{s}^2/2)$$

- ▶ F_t : forecast of $RV_{t+1:t+h}$ made at t ; $\hat{y}_t$: fitted log forecast; $\hat{s}^2$: variance of the regression residuals
- ▶ $+\hat{s}^2/2$: since $\mathbb{E}[e^Z] = e^{\mu+s^2/2}$ for $Z \sim N(\mu, s^2)$; exact if the log error is conditionally Normal with constant variance, otherwise an approximation

- For $h > 1$ targets overlap: this induces serial dependence (through lag $h - 1$ by construction, possibly longer), so every test uses a HAC (heteroskedasticity and autocorrelation consistent) variance

Targets and Losses (2/2): Robust Losses

- A loss is **robust** if, for a conditionally unbiased proxy, the *expected*-loss ranking with RV_t equals that with the true variance Patton (2011)
 - ▶ Conditionally unbiased: $\mathbb{E}_{t-1}[RV_t] = \sigma_t^2$, with $\mathbb{E}_{t-1}$ the expectation given information up to $t - 1$ and σ_t^2 the true variance of day t
- The two robust losses used here (written for $h = 1$; for $h > 1$ replace RV_t by the target)

$$MSE_t = (RV_t - F_t)^2, \quad QLIKE_t = \frac{RV_t}{F_t} - \ln \frac{RV_t}{F_t} - 1 \geq 0$$

- ▶ F_t : the forecast of RV_t ; QLIKE is zero only if $F_t = RV_t$ and grows faster when the forecast is too low than when it is too high
- QLIKE depends on the ratio RV_t/F_t : the same proportional error has the same loss on a calm day as on a stormy day
- MSE is dominated by a few crisis days; QLIKE is the primary loss, MSE the secondary one; a model is compared through its mean loss over the evaluation days

VOLARE: an Open Archive of Realised Measures

- Built at the Universities of Messina and Florence from tick data [Cipollini et al. \(2026\)](#)
 - ▶ Asset-specific cleaning, trading calendars and timestamp precision; the same estimators for every asset class
- Free for research at volare.unime.it; not redistributed with this course
 - ▶ Download the realised-variance files and point the notebooks to your local folder
- Citation requested by the authors
 - ▶ Cipollini, Cruciani, Gallo, Insana, Otranto & Spagnolo (2026), arXiv:2602.19732



University of Messina, the home of VOLARE

Photo: [Superchilum \(2013\)](#); CC BY-SA 3.0; [Wikimedia Commons](#)

The Panel: 50 Assets in Three Classes

Class	Assets	Sample	Observations	Median volatility
American large-cap stocks	40	2 Jan 2015 – 30 Jan 2026	111 440	17.1%
Futures (C, CL, ES, GC, NG)	5	28 Sep 2009 – 30 Jan 2026	21 059	19.7%
FX (AUD, EUR, GBP, CAD, JPY vs USD)	5	25 Sep 2009 – 30 Jan 2026	21 203	7.3%

▣ Assets

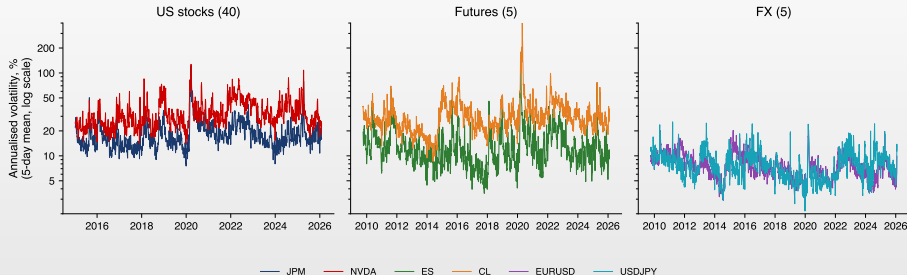
- ▶ Stocks: 40 members of the Dow Jones 30 and S&P 100 (AAPL to XOM)
- ▶ Futures: corn, crude oil, E-mini S&P 500, gold, natural gas

▣ Median volatility: $\sqrt{252 \text{ median}(RV_t)}$ per asset, then the median over the class

- ▶ 252 trading days a year turn a daily variance into an annual one; the data contain trading days only

▣ Variables: five-minute RV, semivariances RV^+ , RV^- , realised quarticity (RQ), daily open and close prices

Realised Volatility Across Asset Classes



- Annualised volatility $\sqrt{252 RV_t}$, five-day mean, log scale, two assets per class (stocks: JPM = JPMorgan Chase, NVDA = Nvidia; futures: ES = E-mini S&P 500, CL = crude oil; FX: EURUSD, USDJPY = US dollar in yen); typical levels: FX about 7.3%, stocks about 17.1% a year
- All classes share March 2020; crude oil (CL) peaks in April 2020, when the front contract traded below zero: a forecaster must adapt within days
- On a log scale volatility is stationary and persistent (the long memory HAR approximates); spikes decay within weeks, so the shape of the recent path carries information

HAR: Three Averages of the Past (1/2)

- HAR (Heterogeneous AutoRegressive) model [Corsi \(2009\)](#): future RV is a linear function of the daily, weekly and monthly averages of past RV

$$RV_{t+1:t+h} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+h}$$

$$RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}, \quad RV_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} RV_{t-j}$$

- Notation

- ▶ $RV_t^{(w)}, RV_t^{(m)}$: weekly (5-day) and monthly (22-day) averages of RV up to day t ; ε_{t+h} : forecast error
- ▶ β_0 : intercept; $\beta_d, \beta_w, \beta_m \geq 0$ in practice: how much yesterday, last week and last month weigh; estimated by OLS for each horizon h
- ▶ $\beta_d + \beta_w + \beta_m$ close to 1: high persistence of volatility

- Traders at daily, weekly and monthly frequencies: a cascade that mimics long memory with three parameters

HAR: Three Averages of the Past (2/2)

- HAR is a restricted AR(22): the 22 lag coefficients take only three distinct values

$$\phi_1 = \beta_d + \frac{\beta_w}{5} + \frac{\beta_m}{22}, \quad \phi_{2..5} = \frac{\beta_w}{5} + \frac{\beta_m}{22}, \quad \phi_{6..22} = \frac{\beta_m}{22}$$

- ▶ ϕ_j : the coefficient of RV_{t+1-j} in an AR(22) (autoregression with 22 lags); writing out the two averages gives these values

- log-HAR: the same regression in logs

$$\ln RV_{t+1:t+h} = \beta_0 + \beta_d \ln RV_t + \beta_w \ln RV_t^{(w)} + \beta_m \ln RV_t^{(m)} + \varepsilon_{t+h}$$

- ▶ No negative forecasts, less driven by spikes; back to levels with $F_t = \exp(\hat{y}_t + \hat{s}^2/2)$
- ▶ Logs of averages are not averages of logs, so this log-HAR is not an exact restricted AR(22) in $\ln RV$ (Seminar 20, A5)

Two Refinements (1/2): HARQ

- RV is a generated regressor: it measures the true variance with an error

$$RV_t = IV_t + \eta_t, \quad \text{Var}(\eta_t) \approx 2 IQ_t / M$$

- η_t : measurement error; errors-in-variables attenuate (pull towards zero) β_d , most on noisy days

- HARQ Bollerslev, Patton & Quaedvlieg (2016): the weight on yesterday's RV falls when RV is measured less precisely

$$RV_{t+1:t+h} = \beta_0 + (\beta_d + \beta_Q(RQ_t^{1/2} - \overline{RQ^{1/2}}))RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+h}$$

- $RQ_t^{1/2}$: square root of realised quarticity, a measure of the error of RV_t ; $\overline{RQ^{1/2}}$: its in-sample mean, so β_d is the weight on an average day
- Expected $\beta_Q < 0$: on a noisy day (RQ_t above average) yesterday's RV gets less weight

- BPQ filter: a forecast outside the in-sample range of the target is replaced by the in-sample mean ("insanity filter")

Two Refinements (2/2): SHAR

- SHAR Patton & Sheppard (2015): split the daily term by the sign of the intraday returns

$$RV_{t+1:t+h} = \beta_0 + \beta^+ RV_t^+ + \beta^- RV_t^- + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+h}$$

- ▶ RV_t^+ , RV_t^- : the positive and negative semivariances of day t ; β^+ , β^- : their weights in the forecast
- ▶ $\beta^+ = \beta^-$ gives back HAR, since $RV_t^+ + RV_t^- = RV_t$
- Patton and Sheppard find $\beta^- > \beta^+$
 - ▶ Variance from down-moves predicts more future variance than variance from up-moves: an intraday version of the leverage effect (Chapter 5)
- The signature can capture a related effect from daily data: the Lévy area between returns and log RV (Path Signatures section)

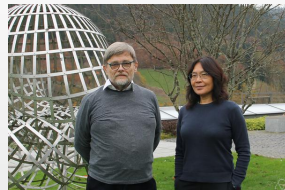
Seven Models in One Table

Model	Target	Weights	Regressors
HAR	RV	equal	$RV_t, RV_t^{(w)}, RV_t^{(m)}$
log-HAR	$\ln RV$	equal	the same, in logs
HARQ	RV	equal	$HAR + RV_t (RQ_t^{1/2} - \overline{RQ^{1/2}})$
SHAR	RV	equal	$RV_t^+, RV_t^-, RV_t^{(w)}, RV_t^{(m)}$
log-HAR-K	$\ln RV$	signature kernel	log-HAR regressors
Sig-L	$\ln RV$	equal	log-HAR + 39 signature terms, two-step LASSO
Sig-LK	$\ln RV$	signature kernel	as Sig-L: the method of Gu et al.

- The last four form a 2×2 ablation (each ingredient switched on and off separately): signature features (yes/no) $\times$ kernel weights (yes/no)
- Log-HAR is the natural baseline for the signature models: same target, and its regressors are among the candidates; but the LASSO may also drop log-HAR terms, so Sig-L vs log-HAR compares two procedures, not only two feature sets

Where Signatures Come From

- **Chen (1957)**: iterated integrals of a path form an algebra and determine it up to reparametrisation
 - ▶ Kuo-Tsai Chen, geometry of loop spaces
- **Lyons (1998)**: rough path theory; the signature is the natural feature set for controlled differential equations
 - ▶ A response to a path depends on it (to first order) only through its signature
- **Machine learning**: [Levin, Lyons & Ni \(2013\)](#), [Chevyrev & Kormilitzin \(2016\)](#), [Lyons & McLeod \(2022\)](#); finance: signature-based asset price models [Cuchiero, Gazzani & Svaluto-Ferro \(2023\)](#)



Terry Lyons (left), Oberwolfach, 2024

Photo: [Katrin Schmid \(2024\)](#); CC BY-SA 2.0 DE; [Wikimedia Commons](#)

A Window of Data Is a Path

- Take the last L days of $d - 1$ series, add time as a channel, and connect the points linearly

$$X : [0, 1] \rightarrow \mathbb{R}^d, \quad X_{u_i} = \left(u_i, \ln \text{RV}_{t-L+1+i}, \sum_{j \leq t-L+1+i} r_j \right), \quad u_i = \frac{i}{L-1}, \quad i = 0, \dots, L-1$$

- ▶ L : window length in days (here 22); d : number of channels (here $d = 3$: time, log RV, cumulative return); $u_i \in [0, 1]$: rescaled time of the i -th point; r_j : open-to-close return of day j
- **Time augmentation**: the first channel is time itself
 - ▶ Without it the signature cannot tell a fast move from a slow one (reparametrisation invariance)
- A regression on the window sees 22 lags; a path summary sees their order, speed and co-movement
 - ▶ Example: RV rose while prices fell, or RV rose first and prices fell later

The Signature: Definition (1/2)

- One signature term is an iterated integral of the path along a word of channels

$$S(X)_{a,b}^{i_1 \dots i_k} = \int_{a < u_1 < \dots < u_k < b} dX_{u_1}^{i_1} \dots dX_{u_k}^{i_k}$$

- Notation

- ▶ $X : [a, b] \rightarrow \mathbb{R}^d$: a path of bounded variation (finite total length), e.g. piecewise linear; X_u^i : its channel i at time u
- ▶ Word $(i_1, \dots, i_k)$: an ordered list of k channels, letters in $\{1, \dots, d\}$; k is the **level** of the term
- ▶ $u_1 < \dots < u_k$: the integral runs over ordered times; dX_u^i : increment of channel i at time u (for a linear segment, slope times du)

The Signature: Definition (2/2)

- The signature is the whole sequence of levels

$$\text{Sig}(X) = (1, S^{(1)}, S^{(2)}, S^{(3)}, \dots), \quad S^{(k)} \in (\mathbb{R}^d)^{\otimes k}$$

- ▶ $S^{(k)}$: the d^k terms of level k , one for each word of length k , stored as a k -dimensional array (a tensor; $\otimes$: tensor product)
- ▶ $\text{Sig}^N(X)$: the truncated signature, levels 0 to N ; N is the depth

- Level 1: the total increments of the channels

$$S^i = X_b^i - X_a^i$$

- Level 2: increments of channel j weighted by how far channel i has already moved

$$S^{ij} = \int_a^b (X_u^i - X_a^i) dX_u^j$$

- ▶ Level 3 adds one more integral; higher levels record finer details of the order of the moves

Linear Segments and Chen's Identity (1/2)

- A linear segment with increment $\Delta \in \mathbb{R}^d$ has the tensor exponential as signature

$$\text{Sig}(\text{segment}) = \exp_{\otimes}(\Delta) = \left(1, \Delta, \frac{1}{2!}\Delta^{\otimes 2}, \frac{1}{3!}\Delta^{\otimes 3}, \dots\right)$$

- ▶ $\Delta = X_b - X_a$: the increment of the segment; $\Delta^{\otimes k}$: the array of all products $\Delta^{i_1} \dots \Delta^{i_k}$, so a level- k term is $\Delta^{i_1} \dots \Delta^{i_k} / k!$

- **Chen's identity** [Chen \(1957\)](#): the signature of two paths run one after the other is the tensor product of their signatures

$$\text{Sig}(X * Y) = \text{Sig}(X) \otimes \text{Sig}(Y), \quad (A \otimes B)^{(k)} = \sum_{p=0}^k A^{(k-p)} \otimes B^{(p)}$$

- ▶ $X * Y$: concatenation, Y starts where X ends; $A^{(k)}, B^{(k)}$: level k of the signatures A, B , with $A^{(0)} = B^{(0)} = 1$
- ▶ Level k of the whole path collects every way of splitting a word of length k between the first and the second piece

Linear Segments and Chen's Identity (2/2)

- ▣ A piecewise-linear path with m segments, increments $\Delta_1, \dots, \Delta_m$

$$\text{Sig} = \exp_{\otimes}(\Delta_1) \otimes \dots \otimes \exp_{\otimes}(\Delta_m)$$

- ▶ This is the whole algorithm: m updates, each costing $O(d^N)$ at depth N (a few lines of numpy); $O(\cdot)$: order of the number of operations

- ▣ Rolling windows: Chen also removes an initial piece

$$\text{Sig}(X_{[s,t]}) = \text{Sig}(X_{[a,s]})^{-1} \otimes \text{Sig}(X_{[a,t]})$$

- ▶ $X_{[s,t]}$: the path restricted to $[s, t]$, with $a \leq s \leq t$; $(\cdot)^{-1}$: inverse for $\otimes$, the signature of the piece run backwards

Worked Example: a Two-Segment Path

Path

- ▣ $X : (0, 0) \rightarrow (1, 0) \rightarrow (1, 2); \Delta_1 = (1, 0), \Delta_2 = (0, 2)$
- ▣ Level 1: $S^{(1)} = \Delta_1 + \Delta_2 = (1, 2)$
- ▣ Level 2 by Chen: $S^{ij} = \frac{1}{2}\Delta_1^i\Delta_1^j + \Delta_1^i\Delta_2^j + \frac{1}{2}\Delta_2^i\Delta_2^j$
- ▣ Δ_m^i : component i of the increment of segment m ; the middle term: channel i moves in segment 1 before channel j moves in segment 2

Result

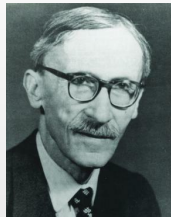
- ▣ $S^{11} = \frac{1}{2}, S^{12} = 0 + 1 \cdot 2 + 0 = 2$
- ▣ $S^{21} = 0, S^{22} = \frac{1}{2} \cdot 2^2 = 2$
- ▣ Check: $S^{11} = (\Delta X^1)^2/2$ and $S^{22} = (\Delta X^2)^2/2$ (shuffle identity)
- ▣ $S^{12} \neq S^{21}$: the order matters (first right, then up)

The Lévy Area: Order as a Number

- The Lévy area A^{ij} : the antisymmetric part of level 2

$$A^{ij} = \frac{1}{2}(S^{ij} - S^{ji})$$

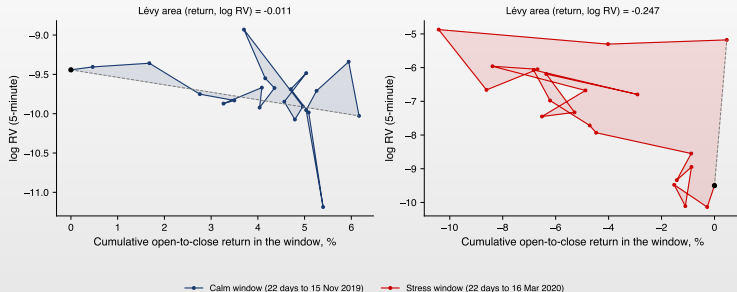
- ▶ Equals the signed area between the path, projected on channels (i, j) , and its chord (the straight line from start to end)
- ▶ $A^{ij} > 0$: channel i tends to move before channel j ; $A^{ij} = -A^{ji}$
- Symmetric part: $\frac{1}{2}(S^{ij} + S^{ji}) = \frac{1}{2} \Delta X^i \Delta X^j$, no new information
 - ▶ Example above: $A^{12} = (2 - 0)/2 = 1$, the area of the triangle under the path
- Channels (return, log RV): a price fall followed by a volatility rise gives $A < 0$, but so does a rise followed by a volatility fall
 - ▶ A lead-lag statistic: leverage-type sequences (Chapter 5) are one contributor to a negative area; the sign alone does not identify them



Paul Lévy (1886–1971)

Photo: Konrad Jacobs; CC BY-SA 2.0 DE;
[Wikimedia Commons](#)

Two Windows of JPM as Paths



- Path (cumulative open-to-close return, log RV) over 22 days; dashed chord from start to end; the shaded region gives the Lévy area
- Calm window (to 15 Nov 2019): log RV moves by -0.58, the path wanders, Lévy area -0.011; stress window (to 16 Mar 2020): log RV rises by +4.32 ($RV \times 75$) while returns fall, Lévy area -0.247
- The net intraday return is only +0.5%: the crash was largely overnight, outside this channel; no signature level recovers what the input omits

Why Signatures Work (1/2): Uniqueness and Shuffles

- **Uniqueness** Hambly & Lyons (2010): the full signature identifies the path
 - ▶ $\text{Sig}(X)$ determines X up to tree-like equivalence (adding pieces that are retraced backwards) and translation
 - ▶ With a strictly increasing time channel nothing can be retraced, so only translation remains
 - ▶ A finite truncation Sig^N is not injective: two different paths can share levels $1..N$
- **Shuffle identity**: a product of signature terms is a sum of higher terms

$$S^i S^j = S^{ij} + S^{ji}, \quad S^{ij} S^k = S^{ijk} + S^{ikj} + S^{kij}$$

- ▶ The sum runs over all interleavings (shuffles) of the two words that keep the order inside each word
- ▶ So a linear model in $\text{Sig}^N(X)$ already contains polynomials of the increments

Why Signatures Work (2/2): Universality and Decay

- **Universality:** any continuous function of the path is approximately linear in the signature (Stone–Weierstrass, via the shuffle identity)

$$\forall \varepsilon > 0 \quad \exists \ell \text{ linear} : \sup_{X \in K} |f(X) - \langle \ell, \text{Sig}(X) \rangle| < \varepsilon$$

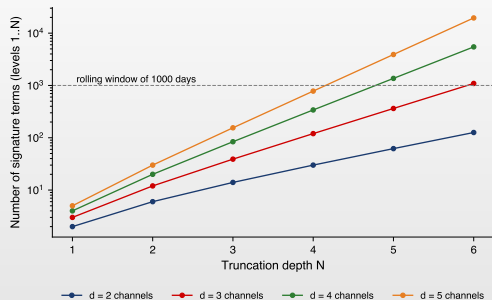
- ▶ K : a compact set of time-augmented paths with a common start (in the 1-variation topology); $f : K \rightarrow \mathbb{R}$: a continuous function of the path
- ▶ $\langle \ell, \text{Sig}(X) \rangle$: a weighted sum of finitely many signature terms, i.e. a linear regression on them
- ▶ This is why a LASSO on signatures can approximate nonlinear functions of the path; the price is d^N features, so sparsity is needed

- **Factorial decay:** high levels are small

$$\|S^{(k)}\| \leq \frac{\|X\|_{\mathbf{1-var}}^k}{k!}$$

- ▶ $\|X\|_{\mathbf{1-var}}$: total length of the path (sum of the lengths of its segments); $\|S^{(k)}\|$: size (Euclidean norm) of level k
- ▶ Truncating at a small depth N loses little for paths of moderate length, which is why channels are rescaled ([▶ Appendix](#))

How Many Features? The Truncation Dimension



- Number of terms in levels 1..N of a d -dimensional path (a geometric sum, since level k has d^k words)
 - ▶ $\dim(d, N) = \sum_{k=1}^N d^k = \frac{d^{N+1}-d}{d-1}$
- This chapter: $d = 3$ channels (time, log RV, cumulative return), depth $N = 3$: $3 + 9 + 27 = 39$ terms
 - ▶ Three are constant (words with only the time letter), so 36 vary
- $d = 5$, $N = 4$ would give 780 terms for 1000 observations: the LASSO would select from noise
- Gu et al. do not disclose N and L ; here $N = 3$, $L = 22$ are fixed ex ante, with $N = 2$ as a robustness check

Two Invariances to Keep in Mind

- ▣ **Translation invariance:** $\text{Sig}(X + c) = \text{Sig}(X)$ for any constant vector $c \in \mathbb{R}^d$: only increments enter
 - ▶ The signature does not see the **level** of volatility, only the shape of its recent path
 - ▶ So the level must enter separately: the log-HAR terms play the role of the external factors x_τ of Gu et al.
- ▣ **Reparametrisation invariance:** running the path faster or slower, $X_{\psi(u)}$ with ψ increasing, leaves Sig unchanged
 - ▶ Removed by the time channel; basepoint augmentation (prepending a segment from 0 to the original starting point) would remove the first invariance [Morrill et al. \(2020\)](#)
 - ▶ Translating the path to start at 0 does not: it is itself a translation (Seminar 20, C1 uses the prepended segment)

The Signature Kernel

- Kernel = inner product of feature maps [Király & Oberhauser \(2019\)](#): a similarity score between two paths

$$k_N(X, Y) = \langle \text{Sig}^N(X), \text{Sig}^N(Y) \rangle = 1 + \sum_{k=1}^N \sum_{|w|=k} S(X)^w S(Y)^w$$

- ▶ X, Y : two paths with the same channels; w : a word, $|w|$: its length (level); $S(X)^w$: the term of word w ; the leading 1 is level 0
- Distance induced by the kernel (Gu et al., Eq. 9): squared Euclidean distance between the truncated signatures

$$d_{\text{Sig}}(X, Y) = k_N(X, X) - 2k_N(X, Y) + k_N(Y, Y) = \|\text{Sig}^N(X) - \text{Sig}^N(Y)\|^2$$

- ▶ $d_{\text{Sig}} = 0$: the two windows have the same signature up to depth N ; large d_{Sig} : different shapes
- Gu et al. and this chapter use this **truncated** kernel, with N as for the features
 - ▶ The untruncated kernel solves a Goursat PDE (partial differential equation) [Salvi et al. \(2021\)](#); the paper cites it but does not use it
- Channels are rescaled before the kernel (▶ [Appendix](#)): otherwise the channel with the largest units dominates

The Problem at Amazon and the Features

- Weekly North American truckload rates, national and five regions, 2018–2022, forecast up to 12 weeks ahead [Gu et al. \(2024\)](#)
 - ▶ More than 100 candidate factors screened to about 40; in Amazon operations since November 2022; KDD 2024 (ACM Conference on Knowledge Discovery and Data Mining)
- Features at week τ (Section 4.1, Eq. 14): external factors x_τ and the signature of the time-augmented target path
 - ▶ $\xi_\tau = [x_\tau, \text{Sig}^N(\hat{y}_{\tau-l:\tau})]$, $\hat{y}_{\tau-l:\tau} = \{(t_i, y_{t_i})\}$
 - ▶ y_{t_i} : the rate in week t_i , $\tau - l \leq t_i \leq \tau$; l : lookback in weeks;
 $\hat{y}_{\tau-l:\tau}$: the time-augmented path of the last l weeks
- One direct regression per horizon Δt (in weeks), $y_{\tau+\Delta t}$ on ξ_τ ; lookback l , depth N , temperature γ (next slides) are not published (confidentiality)



A container truck on an American interstate, 2025

Photo: [DanTD \(2025\)](#); CC BY 4.0; [Wikimedia Commons](#)

The Method: Two-Step LASSO

- Step 1, selection by a weighted LASSO (Least Absolute Shrinkage and Selection Operator) [Tibshirani \(1996\)](#) (Eq. 14)

$$\hat{\theta}_L = \arg \min_{\theta} \sum_{\tau} w_{\tau} (y_{\tau+\Delta t} - \xi_{\tau}^{\top} \theta)^2 + \lambda \|\theta\|_1$$

- θ : coefficient vector; $w_{\tau} \geq 0$: weight of week τ (equal, or from the kernel, next slide); $\|\theta\|_1 = \sum_j |\theta_j|$
- $\lambda \geq 0$: penalty; a larger λ sets more coefficients exactly to zero; $\lambda = 0$ is weighted OLS

- Step 2, refit by weighted OLS on the selected support (Eq. 15)

$$\hat{\theta} = \arg \min_{\text{supp}(\theta)=\text{supp}(\hat{\theta}_L)} \sum_{\tau} w_{\tau} (y_{\tau+\Delta t} - \xi_{\tau}^{\top} \theta)^2$$

- $\text{supp}(\theta)$: the support, the set of nonzero coefficients; only the regressors kept in step 1 are re-estimated
- Why refit: LASSO shrinks the kept coefficients towards zero; OLS after selection removes that bias [Belloni & Chernozhukov \(2013\)](#)
 - The paper calls it “adaptive” because of the weights w_{τ} , not in the sense of the adaptive LASSO of [Zou \(2006\)](#)
- How λ is chosen is not reported

The Method: Kernel Weights

- Distance of each past window to the last window with an observed target (Eq. 11)

$$\delta_{\tau} = d_{\text{Sig}, N}(z_{\tau-l:\tau}, z_{t-\Delta t-l:t-\Delta t})$$

- $z_{\tau} = (x_{\tau}, y_{\tau+\Delta t})$: the *pairs* of predictors and later target, not the features ξ_{τ} (Algorithm 1); $z_{\tau-l:\tau}$: their path over l weeks
- t : the current week; $t - \Delta t$: the last week whose target y_t is already observed; $d_{\text{Sig}, N}$: the signature distance at depth N

- Softmax weights with temperature $\gamma \geq 0$ (Eq. 12): similar windows (small δ_{τ}) get more weight

$$w_{\tau} = \frac{\exp(-\gamma \delta_{\tau})}{\sum_s \exp(-\gamma \delta_s)}$$

- s : runs over all training weeks, so the weights sum to 1; $\gamma = 0$: ordinary (equal-weight) regression; $\gamma \rightarrow \infty$: only the most similar past window
- A nearest-neighbour idea inside a regression: learn from the past periods that looked like today
 - Cost of the idea: fewer effective observations, $n_{\text{eff}} = 1 / \sum_{\tau} w_{\tau}^2$ (Kish); $n_{\text{eff}} = n$ for equal weights $1/n$, $n_{\text{eff}} = 1$ if one week takes all the weight

What the Paper Reports, Read Critically

- National rates, three months ahead, April–November 2021 (Table 1): relative error $|y - \hat{y}|/|y|$ (y : realised rate, $\hat{y}$: its forecast) of about 1–2% vs 13–27% for the industry forecast, “more than fivefold”
 - ▶ Horizons 1–12 weeks (Table 2): 1.06–2.60% nationally, 1.05–5.69% by region; estimated annual saving of about 50 million dollars
 - ▶ Kernel ablation (Table 3, six weekly values, 31 October–5 December 2021, one region): errors up to 9.85% without the kernel vs up to 5.15% with it
- Limits of the evidence
 - ▶ **One benchmark:** an unnamed commercial forecast; no ARIMA, no LASSO without signatures, no random walk
 - ▶ **Short evaluation:** a few months, a few series, no standard errors, no test of equal accuracy
 - ▶ **Undisclosed tuning:** l , N , γ , λ and how they were chosen
- None of this makes the method wrong; it makes the evidence thin, which is why it is tested elsewhere, replication before extension ([Chapter 19](#))

What Transfers to Volatility, and What Does Not

Transfers

- ▣ Signature of the time-augmented recent path of the target
- ▣ Two-step LASSO with weights, direct regression per horizon
- ▣ Softmax weights $\exp(-\gamma \delta_\tau)$ with the truncated signature kernel

Must be adapted

- ▣ Daily, noisy target instead of weekly rates: logs, HAC inference, robust losses
- ▣ External factors x_τ replaced by the log-HAR terms (they carry the level)
- ▣ Undisclosed choices (l, N, γ, λ) fixed ex ante in this chapter
- ▣ Kernel: the path of this chapter (time, ln RV, return) ending at τ vs the window ending at t , updated every 5 days, instead of the pairs $(x_\tau, y_{\tau+\Delta t})$ ending at $t - \Delta t$: an adaptation
- ▣ Strong benchmarks and formal tests instead of one commercial forecast

Hypotheses

- ▣ **H1 (features):** signature terms add information beyond log-HAR
 - ▶ Test: Sig-L vs log-HAR, same target, same equal weights
- ▣ **H2 (weights):** kernel weights improve a given regression
 - ▶ Tests: Sig-LK vs Sig-L and log-HAR-K vs log-HAR
- ▣ **H3 (method):** the full method of Gu et al. beats the HAR family
 - ▶ Tests: Sig-LK vs HAR and vs log-HAR; model confidence set of all seven models
- ▣ **H4 (stress):** any gain is larger in stress periods (COVID-19 crash, 2–30 April 2020, the US tariff shock), where similar past windows should matter most

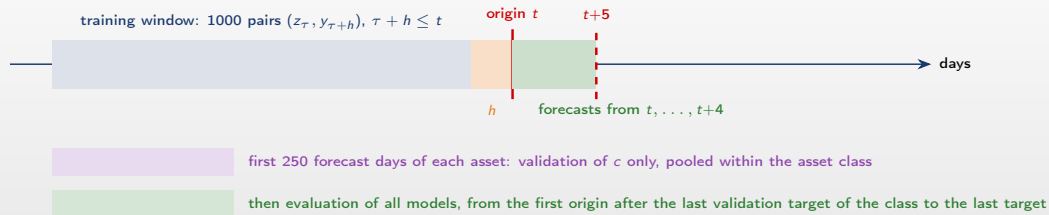
Kernel Temperature: One Choice per Class, on a Validation Block

- Scale-free temperature, set in every training window

$$\gamma = \frac{c}{\text{median}_{\tau}(\delta_{\tau})}$$

- ▶ δ_{τ} : kernel distances of the training days to the current window; $c > 0$: a dimensionless constant, the only tuning parameter
- ▶ Dividing by the median removes the units of δ_{τ} ; $c = 1$: a window at the median distance gets weight e^{-1} relative to an identical one; larger c : more concentrated weights
- Grid $c \in \{0.5, 1, 2, 4\}$, chosen per horizon and asset class on the first 250 forecast days of each asset, pooled within the class
 - ▶ Criterion: mean over the class of $\text{QLIKE}(c)$ divided by the asset's average over the grid
 - ▶ Evaluation starts after the last validation target of the class, so no tuning target postdates an evaluated origin
- For the 40 stocks the smallest temperature, $c = 0.5$ (the most nearly equal weights), wins at every horizon for Sig-LK; for futures and FX the choice changes with the horizon: a flat, noisy criterion (all values in the [Appendix](#))

Rolling Estimation, Step by Step



- Re-estimate every 5 trading days on the last 1000 complete pairs; between re-estimations the coefficients are fixed but the regressors are updated daily
- Kernel distances are recomputed at each re-estimation, against the window ending at the origin t

Evaluation Protocol (1/2): the DM Test

- Per asset and horizon: mean QLIKE and MSE, compared by the DM (Diebold–Mariano) test of equal predictive accuracy [Diebold & Mariano \(1995\)](#)

$$d_t = L(RV_t, F_t^A) - L(RV_t, F_t^B), \quad \text{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}/T}}$$

- ▶ L : the loss (QLIKE or MSE); F_t^A, F_t^B : forecasts of models A and B ; d_t : loss differential; $\bar{d}$: its mean over the T evaluation days
- ▶ $\hat{\Omega}$: Newey–West (HAC) long-run variance of d_t , bandwidth $\max(h, \lceil T^{1/3} \rceil)$ lags; $\lceil \cdot \rceil$: rounding up
- Reading: under H_0 : $\mathbb{E}[d_t] = 0$, DM is approximately $N(0, 1)$
 - ▶ $\text{DM} < -1.645$: model A significantly more accurate at 5% (one-sided); $\text{DM} > 1.645$: significantly less accurate
 - ▶ Rolling estimation, so the test is read as the unconditional GW (Giacomini–White) test; the conditional GW test also asks whether d_{t-h} predicts d_t (instruments $(1, d_{t-h})$) [Giacomini & White \(2006\)](#)

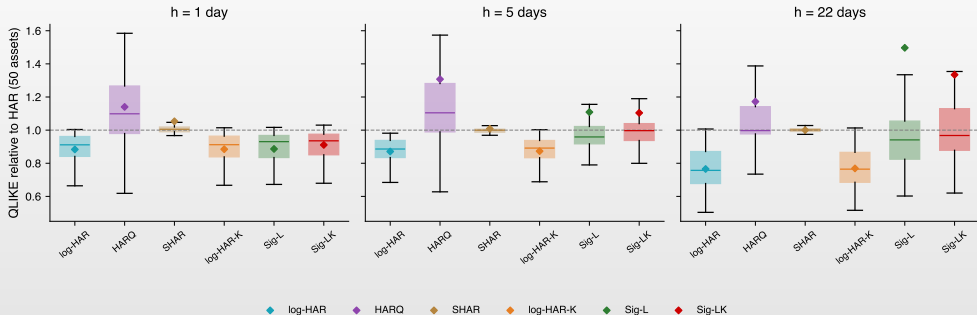
Evaluation Protocol (2/2): Many Models, Many Assets

- MCS (model confidence set) at 90% [Hansen, Lunde & Nason \(2011\)](#): the set of models that contains the best one with probability at least 90%
 - ▶ Built by elimination: while the $T_{\max}$ statistic (the largest standardised gap between a model's mean loss and the average over the models still in the set) rejects equal accuracy, drop the worst model
 - ▶ p -values by the stationary bootstrap, 1000 replications, mean block length $\max(h, 10)$ days
- Panel: DM test on the cross-sectional mean loss differential (one series per date)
 - ▶ Counting wins over 50 assets: the Holm correction controls the FWER (family-wise error rate, the probability of at least one false rejection), BH [Benjamini & Hochberg \(1995\)](#) the FDR (false discovery rate, the expected share of false rejections among the rejections)
- Subperiods: 20 Feb–30 Apr 2020, 2–30 Apr 2025, all other days; the paper's relative error as a descriptive statistic
- 265 929 out-of-sample forecasts per model; stocks are evaluated from January 2020 (the COVID-19 crash is in every stock sample), futures and FX from 2014 (dates in the [Appendix](#))

A Disclosed Deviation

- Pre-registered: the BPQ insanity filter on all models
 - ▶ A test run of the code on JPM (one asset, before any panel result) showed it replacing correct crash forecasts by the calm in-sample mean
 - ▶ On 12 March 2020 the realised next-day RV of JPM was about 40 times the filtered HAR forecast
- Changed before estimating the models on the 50-asset panel: primary results with positivity only (a forecast ≤ 0 becomes the in-sample mean)
 - ▶ The BPQ filter is kept as a robustness check and reported in full
- Rule: a deviation is allowed, a hidden deviation is not; state what was seen when it was changed

QLIKE Relative to HAR, Asset by Asset (1/2)



Ratio for each asset: mean QLIKE of the model over mean QLIKE of HAR; below 1: more accurate than HAR

► Boxes: interquartile range over the 50 assets; line: median; diamond: mean; whiskers: 1.5 interquartile ranges

QLIKE Relative to HAR, Asset by Asset (2/2)

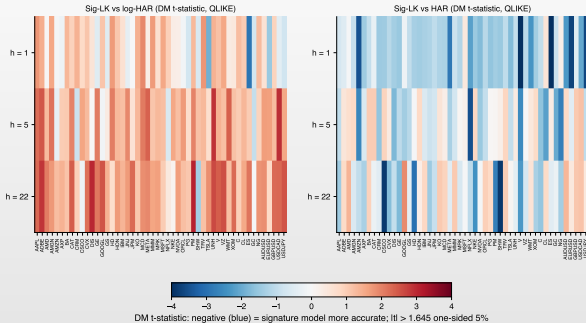
- $h = 1$: log-HAR, log-HAR-K and Sig-L are close (0.883, 0.885, 0.886)
 - ▶ Sig-LK is +4.5% relative to log-HAR: no gain from 36 extra features
- $h = 5, 22$: Sig-LK falls behind log-HAR (+28.0%, +92.9%)
 - ▶ Its medians (0.997, 0.968) are much better than its means: a few assets and days drive the loss
- HARQ and SHAR lose to HAR in levels here
 - ▶ The main design does not apply the BPQ filter (▶ **Appendix**: the filter rescues HARQ)
- First verdict: the method does not beat the simplest log model
 - ▶ A near-tie is not a proof of equal accuracy

Asset by Asset: Wins and Losses, Corrected

Comparison (QLIKE)	$h = 1$	$h = 5$	$h = 22$
Sig-LK vs log-HAR	2/0/0 5/0/0	0/0/0 15/0/2	0/0/0 27/1/16
Sig-L vs log-HAR	3/1/1 5/0/0	0/0/0 9/0/0	1/0/0 21/0/0
log-HAR-K vs log-HAR	3/0/0 3/0/0	3/0/0 1/0/0	5/0/0 6/0/0
Sig-LK vs Sig-L	2/0/0 5/0/0	0/0/0 5/0/0	2/0/0 10/0/0
Sig-LK vs HAR	12/5/6 0/0/0	7/1/1 0/0/0	8/2/4 1/0/0
log-HAR vs HAR	21/5/17 0/0/0	23/13/20 0/0/0	42/25/42 0/0/0

- Cell: significantly better on raw/Holm/BH assets | significantly worse on raw/Holm/BH assets; one-sided DM at 5%, family of 50 assets; Holm controls the FWER under any dependence, BH the FDR under independence or PRDS (positive regression dependence on a subset) [Benjamini & Yekutieli \(2001\)](#)
- H1 (signatures): Sig-L beats log-HAR on 3 of 50 assets at $h = 1$ raw, 1 after Holm; about 2.5 raw wins are expected by chance
- H2 (weights): log-HAR-K vs log-HAR and Sig-LK vs Sig-L: no Holm-significant difference at any horizon
- Against HAR at $h = 1$, Sig-LK wins on 12 assets, but log-HAR alone wins on 21: choose the benchmark that isolates the idea

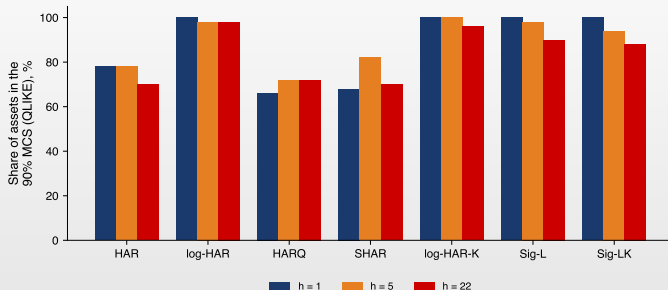
DM Statistics for Every Asset and Horizon



- Diebold–Mariano t -statistics of the QLIKE differential; blue: Sig-LK more accurate, red: less accurate
- Against log-HAR (left): pale at $h = 1$ (blue exceptions ES, TSLA, EURUSD; five assets significantly red before correction), red at $h = 5, 22$
- Against HAR (right): blue at $h = 1$ because Sig-LK inherits the log target; one cell such as ES at $h = 1$ would make signatures look like a clear success



Model Confidence Sets



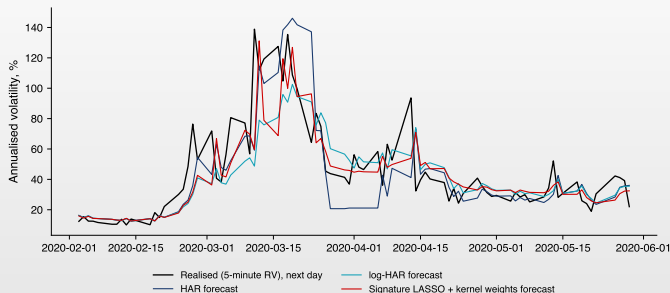
- Share of the 50 assets in which each model belongs to the 90% MCS under QLIKE: log-HAR 100%, 98%, 98%; Sig-LK 100%, 94%, 88% ($h = 1, 5, 22$)
- The MCS rarely separates the four log models; HAR, HARQ and SHAR are excluded for a fifth to a third of the assets: the level target is the clearest loser
- Under MSE almost every model stays in the set [Patton \(2011\)](#): small differences in average loss need very long samples [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)

Panel Tests: Pooling the Evidence

Comparison: DM t (one-sided p)	$h = 1$	$h = 5$	$h = 22$
Sig-LK vs log-HAR	+1.45 (0.927)	+1.68 (0.953)	+1.80 (0.964)
Sig-L vs log-HAR	-1.91 (0.028)	+1.48 (0.931)	+1.51 (0.935)
log-HAR-K vs log-HAR	+1.89 (0.971)	+0.40 (0.654)	+1.92 (0.973)
Sig-LK vs Sig-L	+1.74 (0.959)	+0.98 (0.837)	-0.77 (0.221)
log-HAR vs HAR	-3.09 (<0.001)	-3.95 (<0.001)	-4.50 (<0.001)

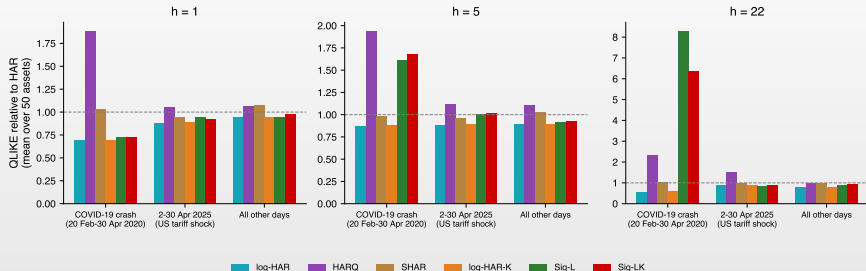
- Series: the mean QLIKE differential over all assets trading that day; negative t = first model better
- Pooled, only the log transform is clearly significant; Sig-L vs log-HAR at $h = 1$ is borderline ($p = 0.028$), driven by futures
 - For the 40 stocks alone the sign flips: $t = +1.72$ against Sig-L
- Conditional GW test, Sig-LK vs log-HAR: 0, 0 and 0 Holm rejections out of 50

Inside the COVID-19 Crash: JPM, Next Day



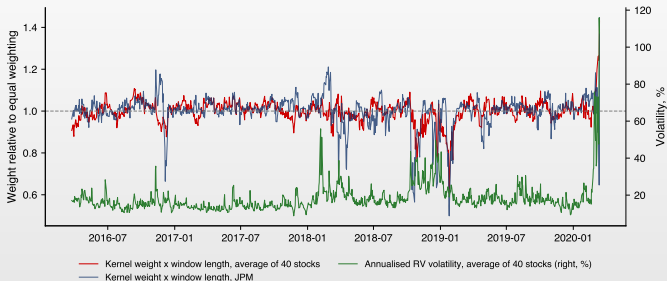
- Annualised volatility $\sqrt{252F_t}$ of the forecasts and $\sqrt{252RV_{t+1}}$ of the target, February–May 2020; all models lag the first jump
- Sig-LK reacts slightly faster than log-HAR in mid-March, when the path has already turned; after the peak both decay at similar speeds
- HAR in levels overshoots, then turns negative in late March: the positivity rule replaces it by the in-sample mean (the flat segment)

Stress Periods vs Normal Times (H4)



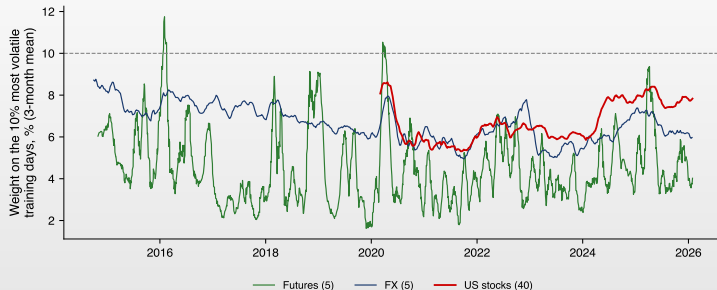
- Mean over assets of QLIKE relative to HAR within each subperiod fixed before the evaluation
- $h = 1$, COVID-19 crash: log-HAR 0.70, Sig-LK 0.73, HARQ 1.89; 2–30 April 2025 (US tariff shock): 0.88 and 0.92; other days: 0.94 and 0.97
- $h = 22$, COVID-19 crash: Sig-LK 6.35 and Sig-L 8.30 times the loss of HAR: exactly where similar past windows should help, the signature models fail most; H4 not supported (descriptive, no formal interaction test)

Kernel Weights at the Origin 16 March 2020



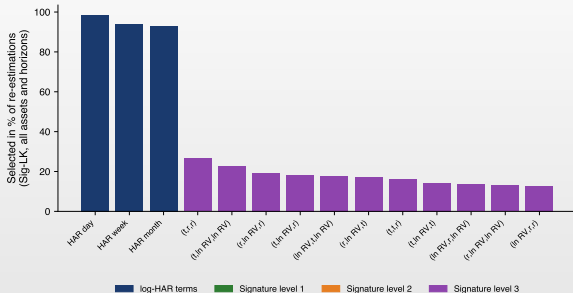
- Weights of Sig-LK ($h = 1$) at the origin 16 March 2020 over the training days (24 Mar 2016 – 13 Mar 2020), times the window length: 1 = equal weighting
- Nearly equal: between 0.61 and 1.43 times $1/n$, effective sample size 975 of 1000 (average over the 40 stocks); the highest weights go to the last days before the origin
- The late-2018 sell-off gets *less* weight; weights are almost uncorrelated with log volatility (+0.04): the translation-invariant signature sees shape, not level

Does the Kernel Ever Favour Stress?



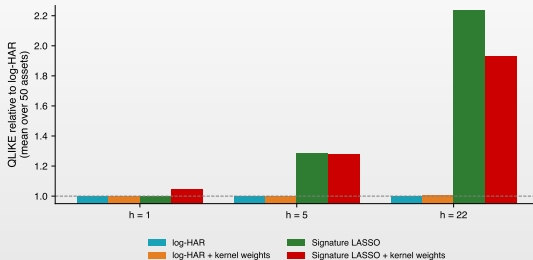
- Weight that Sig-LK ($h = 1$) puts on the 10% most volatile days of its training window, averaged by class, three-month mean
- Stocks: mean 6.7%, between 4.4% and 9.9% (maximum on 17 Mar 2020): always below the 10% of equal weighting; futures and FX show the same pattern
- The mechanism behind the null result for H2: the weights do not implement “learn from past crises”

What the LASSO Selects



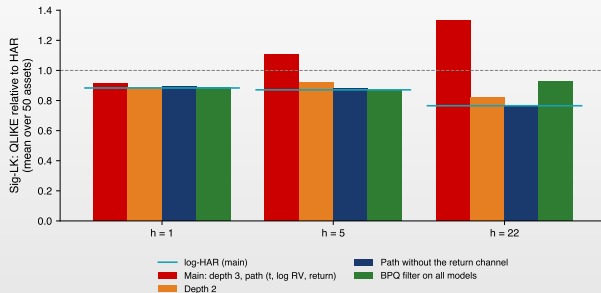
- Share of re-estimations in which a regressor is in the BIC-selected support of Sig-LK; t = time, $\ln RV = \ln RV$, r = cumulative return
- The log-HAR terms are kept in 99%, 94% and 93% of the re-estimations: the LASSO rediscovers HAR; average support 8.7 regressors (Sig-L), 6.7 (Sig-LK)
- The most frequent signature term, (t,r,r) (27%), and all frequent ones are of level 3: cubic in the increments, far outside the training range in a crash, amplified by $\exp(\cdot)$

The 2×2 Ablation



- Mean over the 50 assets of QLIKE relative to log-HAR: features (signature or not) crossed with weights (kernel or equal)
- Weights alone (log-HAR-K): 1.002, 1.002, 1.004: no effect
 - ▶ Average effective sample size 816 of 1000; features alone (Sig-L): 1.004, 1.285, 2.238
- Both (Sig-LK: 1.045, 1.280, 1.929) damp the one-month failure without removing it
 - ▶ They worsen $h = 1$ through crude oil (CL: 2.71); Gu et al. (Table 3) saw a kernel gain in six weekly values of one region

Robustness of Sig-LK



- Sig-LK relative to HAR (mean over 50 assets) in the main design, in two pre-registered variants and with the BPQ filter (the disclosed deviation); the line marks log-HAR
- Depth 2: 0.922 and 0.820 at $h = 5, 22$ (main: 1.104, 1.334); path without returns: 0.878 and 0.770: the long-horizon failures come from level-3 terms with the return channel
- The stable variants only match log-HAR (0.871, 0.765): the null result for H1 does not depend on depth or channels, the instability does

By Asset Class

Sig-L relative to log-HAR (QLIKE)	$h = 1$	$h = 5$	$h = 22$
Stocks (40)	1.009	1.332	2.505
Futures (5)	0.965	1.084	1.206
FX (5)	0.996	1.111	1.134

- The only gain: futures at $h = 1$ (-3.5%) and, marginally, FX
 - Five futures and one horizon: a candidate for a new, pre-registered study, not a finding of this one
- Stocks, 80% of the panel, show no gain: the pooled result is mostly a stock result

Limitations and Open Directions

Limitations of this study

- One design out of many ($L = 22$, $N = 3$, three channels, BIC, a four-point grid for c): a null result for this design is not one for signatures in general
- Truncated kernel only; no external factors (implied volatility, macro news), central in Gu et al.
- Realized GARCH [Hansen, Huang & Shek \(2012\)](#) and machine-learning benchmarks ([Chapter 13](#)) are outside the comparison; stocks are evaluated over six years only

Open directions

- A level-aware kernel: basepoint augmentation or the level of $\ln RV$ in the distance (Seminar 20, C1)
- Signatures of the intraday path (5-minute returns); the PDE kernel [Salvi et al. \(2021\)](#); log-signatures and bounded (tanh) channels against explosive level-3 terms [Morrill et al. \(2020\)](#)
- Cross-asset paths, where Lévy areas measure lead-lag across markets; futures at $h = 1$ on a longer panel, with this design fixed in advance

An Open Question

When does a method transfer across domains?

- Methods travel faster than evidence: a KDD paper on freight rates becomes a candidate for volatility, credit, energy
 - ▶ Which properties of the source problem made the method work?
 - ▶ Does the target share them?
- Finance adds a hard filter: 164 teams on the same data disagree more than their standard errors suggest [Menkveld et al. \(2024\)](#)
- Formal version: a transfer succeeds if the gain over the strongest simple benchmark survives a pre-registered set $\mathcal{S}$ of designs, assets and horizons, with the family corrected
 - ▶ In this chapter: $\mathcal{S} = 50 \text{ assets} \times 3 \text{ horizons} \times 4 \text{ variants}$; Holm/BH were applied per comparison and horizon (families of 50 assets), variants reported descriptively; no joint correction over all 600 cells; the gain did not survive

The Discovery Loop Applied to This Chapter

- **Read:** an assistant lists every equation, hyperparameter and benchmark of the paper and marks what is not disclosed; each item is checked against the PDF
 - ▶ **Map:** for each component, does it transfer, what must change, which benchmark isolates it
- **Pre-register:** hypotheses, grid, tests and correction fixed in a dated document before the first result; the assistant drafts, the team decides every number
- **Build and test the pieces:** signatures against identities (level 1 = increment, Chen on a split path), not against another library
- **Run the grid, then attack it**
 - ▶ which conclusions survive Holm over 50 assets?
 - ▶ which single asset would make a misleading headline?
- **Explain the null:** here flat weights (translation invariance) and explosive level-3 terms (exponential link), both checked with data; the full prompts and the researcher's checklist are in the [▶ Appendix](#)

Mini-Case: Three Headlines From the Same Runs

- “Signatures beat HAR on 12 of 50 assets” (one-day horizon, raw DM)
 - ▶ True, but log-HAR alone beats HAR on 21: the gain belongs to the log transform
- “Signatures improve E-mini S&P 500 volatility forecasts” (ES, $h = 1$, DM $t = -2.91$ against log-HAR)
 - ▶ One cell out of 150 asset-horizon cells: after Holm, 0 of 50 at $h = 1$ and none at $h = 5, 22$
- The headline of the pre-registered evaluation: “no gain over log-HAR; worse at longer horizons; kernel weights inert”
 - ▶ An AI assistant asked to “find where the method works” will produce the first two; the plan protects the third

Your Turn

- Pick one open direction of this chapter (level-aware kernel, intraday signatures, log-signatures, futures only)
 - ▶ Write H_0/H_1 , the benchmark that isolates it, and the result that would make you drop it
- Use an assistant to list what could go wrong in your design, then write one unit test per item
- Fix the pre-registration in a dated document before the first out-of-sample number; run the grid; report the distribution over $\mathcal{S}$
- Log every AI error you catch in `AI_ERRORS.md` (Seminar 20, C2 gives typical ones)

Key Takeaways

- Signatures turn a window into a feature vector with a theory behind it: Chen's identity, uniqueness, universality
- Transferred to realised variance, the method of Gu et al. does not beat log-HAR
 - ▶ H1 (signatures add information): not supported, no gain at one day, a loss at a week and a month
 - ▶ H2 (kernel weights help): not supported, weights are almost flat and down-weight past stress; H3: only HAR in levels is beaten, through the log target; H4 (gains in stress): not supported
- The translation-invariant kernel sees shape, not level; level-3 terms explode in crashes
- The strongest effect in the study is the oldest one: model the log of RV
- A pre-registered null result, with ablations and a mechanism, is a finding

Further Study

Reading

- ▣ Signatures: Chevyrev & Kormilitzin (2016); Lyons & McLeod (2022); realised measures: Franke, Härdle & Hafner (2019) and Chapter 9

Advanced courses on quantinar.com

- ▣  Statistics of Financial Markets
- ▣  Applied Time Series Analysis with Python
- ▣  Machine Learning in Financial Risk

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_20; data: VOLARE, free for research at volare.unime.it

Appendix: Features and Scaling

- Regressors at day τ for the signature models: 3 log-HAR terms + 39 signature terms
 - ▶ $\xi_\tau = [\ln \text{RV}_\tau, \ln \text{RV}_\tau^{(w)}, \ln \text{RV}_\tau^{(m)}, \text{Sig}^3(X_{\tau-21:\tau})]$, path $X = (\text{time}, \ln \text{RV}, \text{cumulative } r)$, 22 points
- Each non-time channel q is multiplied by $\kappa_q = 1/(s_q\sqrt{L-1})$, so its typical total move over the window is about 1
 - ▶ s_q : standard deviation of the daily increments of channel q in the training window up to the origin t ; $L-1 = 21$ increments per window
 - ▶ Time is not rescaled (constant increments, $s = 0$): it runs from 0 to 1; a word $(i_1..i_k)$ is rescaled by $\prod_j \kappa_{i_j}$, so signatures are computed once
- Before the LASSO every column is standardised with weighted moments of the training window; constant columns (words of time only) are dropped
- Raw features at τ use data up to τ ; scales and standardisation use data up to the origin t : the no-look-ahead cutoff is t

Appendix: Estimation: Choosing λ Without Look-Ahead

- Inside each training window: the whole weighted LASSO path by LARS (least-angle regression)
 - ▶ For every support on the path: weighted OLS refit (step 2 of Gu et al.)
- Pick the support that minimises a BIC-type criterion (Bayesian information criterion) with the Kish effective sample size, a heuristic of this design
 - ▶ $\text{BIC}(S) = n_{\text{eff}} \ln \hat{s}_S^2 + (|S| + 1) \ln n_{\text{eff}}$, $n_{\text{eff}} = 1 / \sum_{\tau} w_{\tau}^2$
 - ▶ S : a support (set of kept regressors), $|S|$: its size, $+1$ for the intercept; $\hat{s}_S^2$: weighted residual variance of the refit on S ; first term: fit, second: penalty per parameter
- Why not cross-validation: 50 assets $\times$ 3 horizons $\times$ hundreds of re-estimations; BIC uses only the window
 - ▶ Kish size corrects for unequal weights, not for serial dependence (overlapping targets, $h > 1$): not a likelihood-based BIC for these data

Appendix: Kernel Temperatures Chosen on the Validation Block

- ▣ Chosen c for Sig-LK ($h = 1, 5, 22$)
 - ▶ stocks 0.5, 0.5, 0.5; futures 2.0, 1.0, 0.5; FX 0.5, 2.0, 0.5
- ▣ Chosen c for log-HAR-K ($h = 1, 5, 22$)
 - ▶ stocks 0.5, 0.5, 1.0; futures 2.0, 1.0, 4.0; FX 4.0, 2.0, 4.0
- ▣ Grid $c \in \{0.5, 1, 2, 4\}$; the smallest temperature (the most nearly equal weights) wins for the stocks at every horizon; for futures and FX the choice changes with the horizon

Appendix: What Was Evaluated

Class	Evaluation starts	Ends	Days per asset (median)
Stocks	27 Jan 2020	29 Jan 2026	1 499
Futures	25 Sep 2014	29 Jan 2026	2 919
FX	27 Aug 2014	29 Jan 2026	2 952

- 265 929 out-of-sample forecasts per model ($50 \text{ assets} \times 3 \text{ horizons}$)
- Stocks start in 2015, so their evaluation begins in January 2020: the COVID-19 crash is in every stock sample
 - Futures and FX start in 2009 and are evaluated from 2014

Appendix: The Paper's Metric on VOLARE

Median relative error $ RV - F /RV$ (%)	$h = 1$	$h = 5$	$h = 22$
HAR	57.2	48.9	51.7
log-HAR	50.4	41.9	38.7
HARQ	55.0	49.0	50.7
SHAR	57.9	49.2	51.8
log-HAR-K	49.7	41.0	38.7
Sig-L	50.1	42.6	42.6
Sig-LK	50.6	43.6	43.7

- Median over assets of the mean relative error: about 40–60%, against 1–2% for freight rates in Gu et al.
- Daily RV is a noisy, heavy-tailed target: the evaluated models have much larger relative errors than those reported for freight rates, without establishing a lower bound on attainable error
 - ▶ Sig-LK forecasts above ten times the in-sample maximum: 8, 29, 36 ($h = 1, 5, 22$), on 10 assets at $h = 5$; log-HAR: none

Appendix: The Insanity Filter, Revisited

QLIKE relative to HAR, BPQ filter on all models	$h = 1$	$h = 5$	$h = 22$
HAR	1.000	1.000	1.000
log-HAR	0.763	0.654	0.778
HARQ	0.995	0.912	1.014
SHAR	1.014	1.003	1.000
log-HAR-K	0.772	0.659	0.779
Sig-L	0.871	0.819	0.904
Sig-LK	0.890	0.867	0.930

- The filter rescues HARQ relative to HAR (0.995 at $h = 1$ instead of 1.140), as BPQ intended
- Sig-LK relative to log-HAR, both filtered: 1.238, 1.424, 1.209 (unfiltered 1.045, 1.280, 1.929): the gap widens at $h = 1, 5$ and narrows at $h = 22$
- Filtered forecasts, $h = 1$: HAR 180, HARQ 140, Sig-LK 143, log-HAR 29 (of 89 893): the filter hits the signature models far more often than log-HAR

Appendix: What the Researcher Must Check

- **The method is the paper's:** kernel, weights, refit, as in the cited equations
 - ▶ Test: every design slide cites an equation, table or section of the paper; undisclosed choices are labelled as choices of the study
- **No look-ahead:** λ , scales, c and features use only data available at the origin
 - ▶ Test: change every value after the origin (for example, multiply future RV by 10); the forecast made at the origin must not change
- **The benchmark isolates the idea**
 - ▶ Test: the 2×2 ablation is reported, not only the full model against HAR
- **Losses and tests are the right way round**
 - ▶ Test: $QLIKE(RV, RV) = 0$ and $QLIKE(RV, F) > 0$ otherwise; the DM variance is HAC with bandwidth at least h

Appendix: The Discovery Loop, Prompts (1/2)

- **Read:** extract exactly what the paper did
 - ▶ “From Gu et al. (2024, KDD), list every equation of the estimator, every hyperparameter and its reported value, and the benchmark of each table. Mark what is not disclosed.”
 - ▶ Check each item against the PDF: equation numbers, table numbers, the kernel actually used
- **Map:** what transfers, what must be adapted
 - ▶ “Our target is daily realised variance. For each component of the method, say whether it transfers, what must change, and which benchmark isolates it.”
- **Pre-register:** hypotheses, grid, tests, correction, fixed in a dated document before the first result
 - ▶ The assistant drafts; the team decides every number and signs the dated pre-registration

Appendix: The Discovery Loop, Prompts (2/2)

- **Build and test the pieces:** signatures against identities, not against another library
 - ▶ “Write unit tests for a depth-3 signature: level 1 equals the increment, the symmetric part of level 2 equals half the outer product, Chen's identity on a split path.”
- **Run the grid, then attack it**
 - ▶ “Here are all 600 runs (50 assets $\times$ 3 horizons $\times$ 4 specifications: main, depth 2, no return channel, BPQ filter). Which conclusions survive Holm over 50 assets? Which single asset or subperiod would make a misleading headline?”
- **Explain the null:** find the mechanism, not an excuse
 - ▶ Here: flat weights (translation invariance) and explosive level-3 terms (exponential link), both checked with data

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